



Figure 1: Testing QEMU after installation



Figure 2: Installing Ubuntu in QEMU

```
$ qemu -m 512 -hda ubuntu.img
```

```
# yum install openvpn
# yum install bridge-utils
```

Figure 3 shows the VM running after booting from the virtual hard disk. The next thing we need to do is set up networking.

QEMU networking

Setting up networking on QEMU is tricky work. Let's make use of the virtual network kernel devices TAP and TUN, which are different from hardware Ethernet devices; TAP and TUN are supported only in the kernel (i.e., only in software). TAP operates at the data-link layer, and TUN at the network layer.

QEMU can use the TAP interface to provide full networking support to the guest operating system. Before this, we need to install the VPN (Virtual Private Network) package on the host machine, and set up a bridge between the host and guest OS. Install the *openvpn* and *bridge-utils* packages using *yum*:

Now, we will create two scripts for *qemu-ifup* and *qemu-ifdown*, as given below:

```
#qemu-ifup
/sbin/ifconfig eth1 down
/sbin/ifconfig eth1 0.0.0.0 promisc up
openvpn --mktun --dev tap0
ifconfig tap0 0.0.0.0 up
brctl addbr br0
brctl addif br0 eth1
brctl addif br0 tap0
brctl stp br0 off
ifconfig br0 10.10.10.2 netmask
255.255.255.0
```

This script will be used to start QEMU networking. In the first line, the Ethernet device is disabled. For the interface to be a part of a network bridge, it must have an IP address of 0.0.0.0, which is what we have done in the second line. In lines 3 and 4, we create and bring up the TAP

RHCVA

(Red Hat Certified Virtualization Administrator)

The future is of Virtualization

@IPSR

Get trained by faculty team headed by **RHCA** Certified Professionals





Produced

40

RHCSS
Highest in Asia Pacific

Special OFFER for **RHCSS RHCVA**

RHCA RHCDS RHCSS RHCVA RHCE

Modules Starting Every Week

Awards

- Winner of Red Hat Asia Pacific International Award 2008, 2009 & 2010
- Winner of Red Hat GLS National Awards 2004, 05, 06, 07 & 08

Faculty & Infrastructure

- 15 member faculty crew led by RHCE, RHCVA, RHCSS, RHCDS, RHCA certified professionals
- Global standard infrastructure offered at our International Centre located in Kochi

Achievements

- Produced 3000+ RHCEs, 35 RHCSSs & 3 RHCA

ipsr solutions ltd
enabling excellence

Kochi South • Kochi North • Kozhikode
Trivandrum • Kottayam

Tel : 0484 - 2344560, 0481 - 2561410 / 20
Mob: 09447294635, 09447169776
www.ipsr.org • training@ipsrsolutions.com